BIOSYSTEMS ENGINEERING 235 (2023) 202–214

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/issn/15375110

Research Paper

Subcutaneous temperature monitoring through ear tag for heat stress detection in dairy cows

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ARTICLE INFO

Article history:
Received 2 May 2023
Received in revised form
20 September 2023
Accepted 3 October 2023

Keywords:
Dairy Cow
Heat stress
Wireless data acquisition
Real-time monitoring
IoT

Heat stress remains one of the greatest threats to the economic viability of the dairy industry. The threat becomes even greater during warm and humid conditions, especially for those larger farms seeking to expand their operations. This study sought to develop and test a reliable and timely method that allows dairy farmers to monitor the body temperatures of their animals, using a compact, lightweight ear tag device, including a wirelessly rechargeable battery. The device is also capable of wirelessly transmitting a cow's subcutaneous temperature in real-time by means of an injectable, passive RFID biosensor. The outcomes showed that subcutaneous ear-base temperature (EBT) can synchronously follow the core body temperature (CBT) while also producing a comparable temperature gradient. That is, on average, the difference between a CBT and an EBT is 0.68 ± 0.35 °C. Based on regression models, the CBT of a cow can be accurately predicted based on EBT, with an average error of around 0.20 °C and as low as 0.11 °C by using a multivariate polynomial regression model, which includes external environmental conditions as well as the cow's health information, including milk production. The outcomes indicate that there is a consistent, predictable time lag between increases in microenvironmental conditions and increases in a cow's body temperature. These findings should enable dairy barn designers to develop and employ automated control-loop systems that utilise physiological feedback to improve management practices aimed at detecting heat stress and mitigating its effects in a timely manner, while minimising energy use.

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1. Introduction

Traditionally, herd health was monitored by humans and based on observational inspections and experience. However, as the size of dairy operations and the number of animals managed per farmer increased, this monitoring approach became ever more time-consuming and labour-intensive. Therefore, accurately diagnosing the health of individual dairy cows has become an essential aspect of modern dairy barn operation, as animal health status directly impacts the production rate and welfare of a dairy cow. A cow's level of

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https://doi.org/10.1016/j.biosystemseng.2023.10.001
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Nomenclature

Roman Letter Symbols

n	Number of Samples
RH	Relative Humidity, %
t	Time, hr
T	Temperature, °C
ΔT	Temperature Difference, °C
T_{db}	Average Dry Bulb Temperature, °C
x_i	Input Values
y	Predicted Values
y_i	True Values
ẏi	True Values
Z_α	Critical Value

Acronyms

CBT	Core Body Temperature, °C
DCC	Days Carrying Calf, day
DIM	Days in Milk, day
DMAVG	Daily Milk Average, kg day^{-1}
DOY	Day of the Year, day
EBT	Ear Base Temperature, °C
IoT	Internet of Things
LR	Linear Regression
PAR	Parity
PR	Polynomial Regression
RMSE	Root Mean Square Error
THI	Temperature-humidity Index
TOD	Time of the Day, hr

Greek Letters

α	Type 1 Error
β	Regression Parameters
δ	Mean Difference
σ_a	Alternative Standard Deviation
σ_0	Null Standard Deviation

heat stress, for example, can negatively impact the cow's lactation, reproductivity, and mortality, to the extent that the cost of the global dairy industry is expected to rise to a magnitude of billions of dollars (Thornton et al., 2022). Accurate and real-time monitoring of a cow's heat stress can minimise production losses by enabling a dairy farmer to detect such health conditions. It is also essential to achieve effective and efficient use of energy and water with real-time control with timely and optimal cooling methods.

Diary producers have used temperature or thermal indices such as temperature humidity index (THI) as an indirect indication of heat stress. Furthermore, a series of advanced thermal indices, including wind speed as an important parameter, have been developed and tested to assess the levels of heat stress imposed on cattle. However, individual cow responses vary widely depending on various parameters such as days in milk, parity, days carrying calf, daily milk production, etc. Wang et al. (2018) comprehensively reviewed these cattle-related thermal indices and concluded that each thermal index is distinct to an extent that it should be selected and employed carefully. On the other hand, respiration rates and body temperature are both direct ways to assess how well cows are managing heat stress in real time. Atkins et al. (2018) developed a sensor system to measure the respiration rate of eight lactating Holstein cows. Variation in continuously measured respiration rate corresponded to changes in THI and core body temperature (CBT). However, the development of a miniaturised, real-time, wireless sensor with a rechargeable battery remains to be a challenge.

Recently, many studies have sought to develop automated methods for monitoring various aspects of a dairy cow's health and behaviour. One study, for example, found that a wearable accelerometer sensor in the form of a collar or ear tag could be used to quantify and estimate a cow's feeding and rumination time, and those parameters could be used to detect respiratory diseases and subacute ruminal acidosis (Wolfger et al., 2015). The accelerometer sensor could also be used to monitor the time a cow spent lying down, standing, and walking (Chapa et al., 2020; Nogoy et al., 2022), all of which are key indicators, as time spent lying down is often associated with the wellness of a cow, and the lameness caused by excessive standing significantly affects a cow's productivity (Cook & Nordlund, 2009). Advancing computer vision technologies can be used effectively to monitor these physical behaviours and feeding habits, and postures that indicate lameness (Jiang et al., 2022; Jorquera-Chavez et al., 2019). Also, the respiration rate can be measured by means of a physical pressure sensor mounted on the abdomen of the animal (Atkins et al., 2018) or by using an acoustic sensor (de Carvalho et al., 2020) or a wave radar-based system (Tuan et al., 2022) to count breaths. It should be also noted that Ji et al. (2020) summarized the various measurement options for heat stress indicators related to environmental and/or animal responses, in addition to thermal indices and mitigation strategies.

In addition to these many monitoring parameters that can be used to assess a cow's health, an animal's body-temperature trend can provide much of the vital information associated with heat stress and disease. That is, an adult Holstein dairy cow's body temperature typically ranges from 37.8 to 39.2 °C, and any temperature that exceeds this range will indicate systemic distress (Burfeind et al., 2012; de Andrade Ferrazza et al., 2017; Vitali et al., 2020). Historically, the cow-body temperature has been monitored by inserting a rectal or vaginal temperature device (Andersson et al., 2016; Wang et al., 2020). However, inserting these devices requires significant manual labour, and accessing in real-time the data they collect is difficult.

Advanced sensing technologies have allowed for many different body-temperature-measurement approaches. Although the anatomical site can influence the body-temperature measurement, a cow's body-temperature trend can be determined based on either internal or external measurements (Taylor et al., 2014). An ingestible biosensor, for example, with wireless communication capability, can be placed inside the cow's rumen to enable real-time monitoring of the rumen temperature and detection of the oestrus cycle as well as such diseases as mastitis (Kim et al., 2019; Tao et al., 2019). However, the measurements are inconsistent due to water drinking events. Also, a thermal infrared camera can be used to measure the surface temperature of the udder or eye to determine temperature trend (Yan et al., 2021). In one of the

most recent studies focused on how best to measure body temperature, an infrared photodiode thermometer was developed and tested for its ability to monitor a cow's temperature in real-time (Murugeswari et al., 2022). Overall, these readings are far from the core body temperature values using vaginal sensors and sensitive to water spray using soakers and changes in microclimate conditions such as air temperature and wind speed.

A subcutaneous temperature sensor can provide a more stable approach to measuring body temperature as it is less influenced by the environmental conditions during heat waves and certain events such as drinking, spray cooling by soakers, etc.; moreover, the data can be accessed more easily than can data collected by a sensor inserted deep into internal organs. Temperature sensors have already been implanted in various locations on cows to test for accuracy. Lee & Seo, 2021, for one example, surgically inserted thermos-loggers into a cow's neck and around the animal's scapula, and in a more recent study, microchip sensors were implanted near the scutulum of a cow's ear and around the necks of bull calves to monitor body temperature (Setser et al., 2020). Although the functionalities were proven effective, these approaches for monitoring body temperature have predominantly involved either an invasive surgical insertion of an entire physical sensing system or an attachment of a sensor on the skin surface, where the temperature readings can be heavily influenced by microclimate conditions.

As the dairy industry becomes more heavily involved in precision livestock farming — which considers an animal's health status as a key aspect for automatically and efficiently managing animal housing (Morrone et al., 2022; Schillings et al., 2021) — the industry will rely more and more on advancements in the electronics and computing technologies related to the "internet of things" (IoT). As Fig. 1 illustrates, implementing, in place of invasive CBT measurement methods, a real-time, wireless approach to measuring EBT should significantly improve accessibility to each animal's physiological condition and, in return, provide opportunities to develop and improve the methods by which a dairy-barn environment is controlled and managed. That is, the design and system engineers can eventually optimise the operation of cooling fans and soakers based on the real-time monitoring parameters to provide maximum cooling efficiency with minimum energy and water consumption. We are convinced that the EBT data, along with microclimate parameters and other significant factors such as cow's milk yield, can play a significant role in closing the control loop for cooling systems in dairy barns using real-time data analytics. Therefore, the primary objective of the present study is to develop and test an accurate and reliable real-time subcutaneous temperature monitoring system to augment the CBT estimation.

Considering these previous study findings, the present study focuses on developing an ear tag system to minimise the invasiveness while improving data quality. A cow's ear base was deemed an optimal location because of its proximity to the location of the tags that are typically attached to the ears of dairy cows. It should be further noted that Chung et al. (2020) had already demonstrated the viability of using ear-base temperature (EBT) in place of core body temperature (CBT); however, in place of a neck collar wearable device with a biosensor, this study relied on a neck collar equipped with a biosensor and an ear tag device that was much lighter, more compact, rechargeable, and energy-efficient than the previous devices, and could also collect data for longer periods. With the development of this better device, the primary objectives of this study are: (i) Prove that EBT can adequately represent the CBT and thus be used to monitor levels of heat stress, (ii) Quantify the relationship between body temperature and environmental conditions, and (iii) Provide regression models

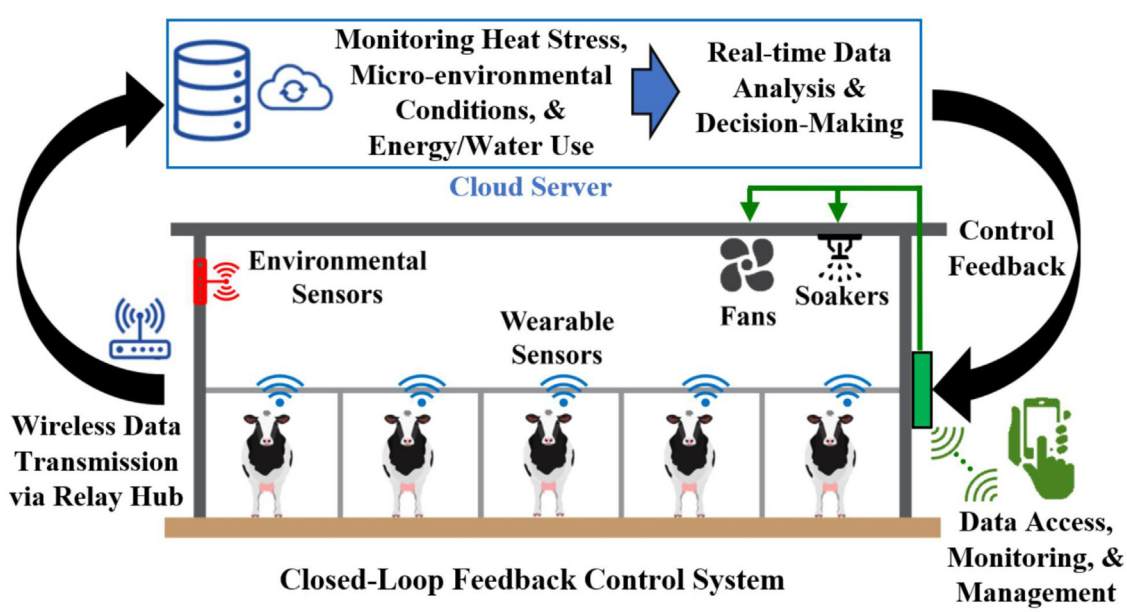


Fig. 1 – A schematic of a closed-loop control system based on dairy cows' physiological and behavioural data and environmental data.

that use EBT to accurately predict CBT by considering environmental conditions and additional health parameters.

2. Materials and methods

2.1. Ear base temperature acquisition

2.1.1. Subcutaneous biosensor placement

The ear-base, subcutaneous temperature of a dairy cow was measured by injecting a passive RFID temperature biosensor (LifeChip Microchip, Destron Fearing Fort Worth, Texas, USA). The biosensor was inserted using a 12-gauge (2.16 mm inner and 2.77 mm outer diameters) injection needle. The backside of the ear base was chosen as the biosensor location because the area contains copious blood vessels from which a temperature can be recorded and enough muscle fibres to hold the biosensor in place. The injection procedure was performed by a trained veterinary surgeon with appropriate preparation of the injection site, one that involved sterilisations and antiseptics, as well as following proper animal-use protocols and standards. The biosensor and the needle were sterilised using ethylene oxide (EO) gas and Steris Applied Sterilisation Technology (Steris PLC, Mentor, Ohio, USA). The injection site was aseptically prepared before the injection procedure, and 3 mL of lidocaine was injected subcutaneously at the base of each cow's ear to ease injection pain.

2.1.2. Wireless ear tag unit and wireless data transmission

With close collaboration, our interdisciplinary team developed a custom wireless ear tag to read EBT from the subcutaneously injected biosensor and transmit the temperature data to a cloud server where users could easily access the data from any internet-connected device in real-time. The primary functions of the wireless ear tag are described in Fig. 2. The ear tag is light and small enough to reside in a cow's ear without causing discomfort. It is circular (11 cm in diameter) and flat in shape and weighs 60 g, and Fig. 3 presents an exploded view of the unit, including all the essential parts. The ear tag was powered by a 150-mAh 3.7-V single-cell Li-ion battery. The assembled tag is waterproof and can reliably read EBT through an RFID from up to approximately 10 cm away which is within

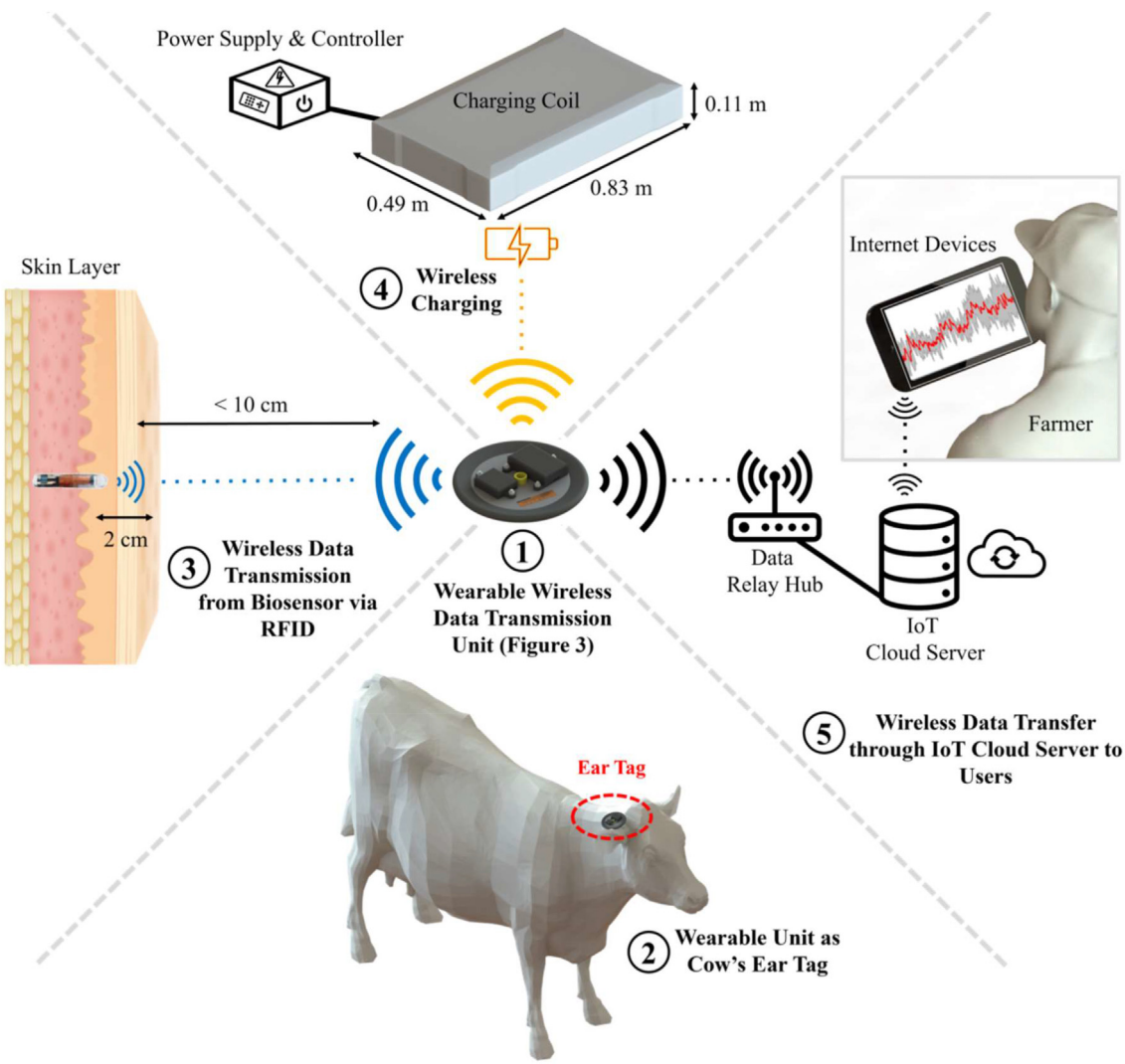


Fig. 2 – Critical elements of the developed ear tag: (1) the ear tag mounted on cow's ear; (2) wireless data transmission from implant sensor via RFID; (3) wireless charging; (4) wireless data transfer through the IoT cloud server to users.

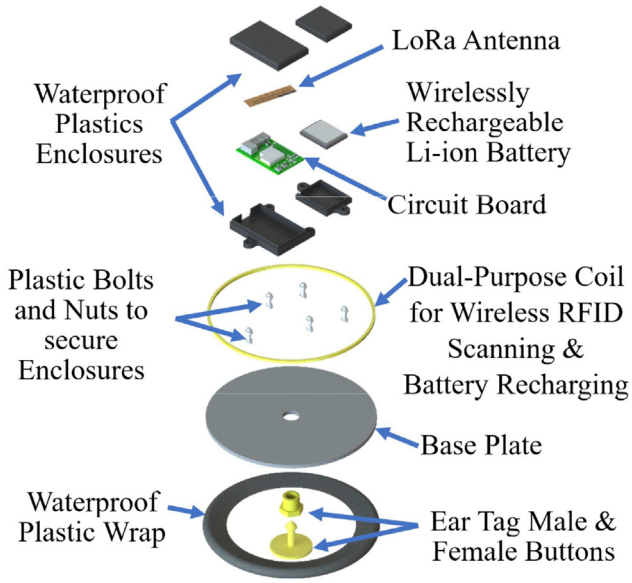


Fig. 3 – An exploded view of the wearable data transmission unit.

range of the implanted biosensor. The ear tag was placed and secured on the top side of cow's ear, which in modern dairy practices is the desirable location for tags other than those used for identification. The ear tag can transmit the EBT data wirelessly to a cloud server by using a combination of LoRa and Wi-Fi communication systems with an additional data relay hub to organise wireless data transmission traffic. Additionally, wireless battery charging was implemented to achieve a lightweight and compact ear-tag design by minimising size and weight, because the battery is usually the heaviest component of any wearable device for its long-term usage.

2.2. *Core body temperature acquisition*

The core body temperature (CBT) was measured to compare it with the subcutaneous temperature (i.e., EBT). The CBT of each cow was measured vaginally using a miniature temperature logger (DST centi-T, Star-Oddi, Gardabaer, Iceland), which has an accuracy of ±0.1 °C, and the logger was shrink-wrapped onto a blank (drug-free) controlled internal drug-release (CIDR) insert (Zoetis, Florham Park, New Jersey, USA) with a 30-s sampling interval. Each packaged vaginal temperature sensor and the vaginal area were sterilised and cleaned with a 10% povidone-iodine solution. At the end of the experiment, all the CBTs were collected by retrieving the sensors.

2.3. *Temperature measurement recalibration*

Before injection, all temperature sensors used to take the CBT and EBT measurements were recalibrated to minimise any reading inconsistency that might be caused by using a different sensor type for each measurement. The sensors were placed in an insulated, constant-temperature water bath. Linear regression was calculated for each sensor against a referencing averaged temperature obtained by means of three thermocouples (T type, Omega Engineering Inc., Stamford, CT, USA) with the temperature readings ranging from 45 °C to 31 °C (±5 °C typical cow's CBT range).

2.4. *Experimental setup*

2.4.1. *Selection of dairy cows*

The temperature data collected for this study were taken from seven randomly selected Holstein dairy cows housed at the Dairy Cattle Centre of the University of Wisconsin–Madison. The details concerning each cow, which were recorded at the start of the experiment, can be found in Table 1. Having different 'days in milk' (DIM), parity (PAR), and other reproduction parameters, as expected the cows in the selected group exhibited a wide range of dairy-cow lactation stages, which in total could reasonably represent a typical herd of cows in a commercial dairy barn. DIM refers to the total number of days that the cow is milking from the calving day, and PAR specifies the number of pregnancies that a cow has undergone. Obtaining data from a group of cows with different production characteristics can minimise statistical bias that can occur because of any similarities between cows, if such exist. All seven cows were free of illness or disease at the outset of the experiment.

2.4.2. *Experimental site*

Figure 4 is a simplified rendering of the indoor experiment facility, including the locations of cows, sensors, and cameras. The experimental facility is a small tie-stall barn within the Dairy Cattle Centre with a highly controlled environment. The enclosed barn is equipped with a negative-pressure ventilation system and evaporative cooling pads, and it can house up to 28 cows. The seven monitored cows were located near the inlets, with three facing south and four facing north, and they were isolated from the rest of the herd with empty stalls immediately adjacent to them. The facility was ventilated by two large (1 m dia.), one medium (0.7 m dia.), and one small (0.5 m dia.) exhaust fans and evaporative cooling pads located at the inlet (1.82 m by 8 m) with wind blowing from west to east.

The fans operated constantly throughout the measurement period, yielding a consistent average wind velocity of 0.95 m/s through the middle alley at a 1.5 m height (around 1.21 m/s at the inlet and 3.51 m/s at the exhaust fans). The wind velocity was measured using a hot-wire anemometer

Table 1 – DIM = Days in milk; PAR = Parity; DCC = Days carrying calf; DMAVG = Daily milk average at the beginning of the experiment.

COW [#]	DIM [days]	PAR [-]	DCC [days]	DMAVG [kg day^{-1}]
1	157	5	74	47.17
2	128	3	0	45.81
3	250	3	172	39.00
4	275	4	158	19.05
5	387	2	165	35.38
6	194	2	67	38.55
7	144	3	32	50.80

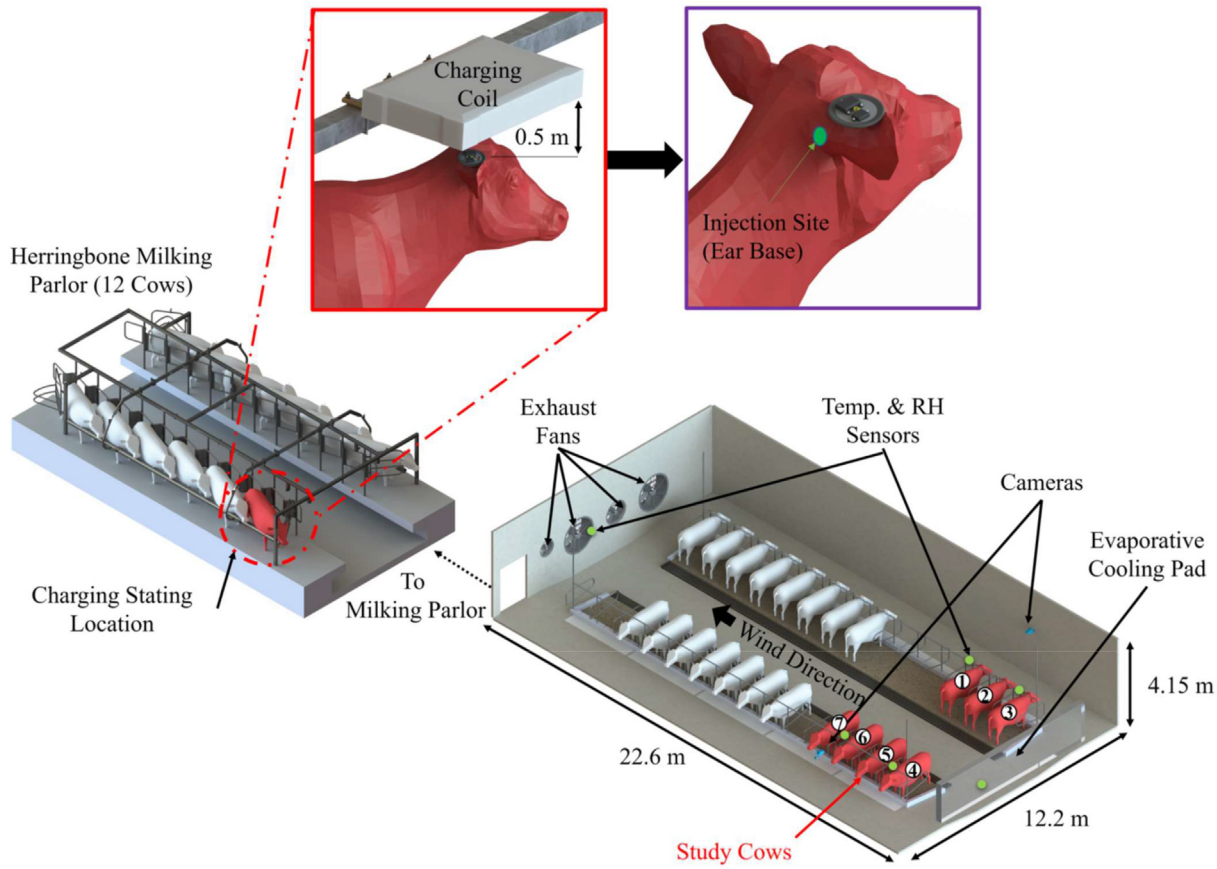


Fig. 4 – Indoor experimental site in the west barn of Dairy Cattle Centre on University of Wisconsin–Madison Campus.

(Alnor Velometer AVM440, TSI Inc., Shoreview, Minnesota, USA). The cows were milked twice a day at around 4 AM and 4 PM, during which times the cows were moved to the milking parlour. After each milking session, the cows were released for about 30 min to the adjacent yard to expose them to the outdoor environment before returning to their stalls. Six micro-environmental sensors (U23-001 HOBO Pro v2 data loggers, Onset Computer Corp., Bourne, Massachusetts, USA), which can measure dry bulb temperature and relative humidity in 1-min sampling intervals, were deployed. The sensors were positioned at carefully designated locations within the barn (marked as a green dot in Fig. 4). During the experiment, two internet protocol cameras (Amcrest 4 MP Bullet wireless IP camera, Amcrest Technology, Houston, Texas, USA) were mounted on the side walls to monitor all seven cows from the front for behavioural anomaly detection.

2.4.3. *Weather and microclimate*

The experiment was conducted from the day of the year (DOY) 231–244 in 2022. Figure 5 summarises the outside weather and indoor THI level that occurred during the experiment. The external weather conditions were surveyed at a location near Dane County Regional Airport Station (43.14069°, −89.34521°). THI, the industry's most used environmental indicator, combines ambient air temperature and humidity level to determine a dairy cow's thermal comfort (Dikmen & Hansen, 2009). The THI was calculated using Equation (1), with averaged air temperature and relative humidity as the dependent parameters obtained from six deployed temperature and relative humidity sensors.

$$THI = (1.8 \times T_{db} + 32) - (0.55 - 0.0055 \times RH) \times (1.8 \times T_{db} - 26.8) \quad (1)$$

where

T_{db} = the average dry bulb temperature, °C

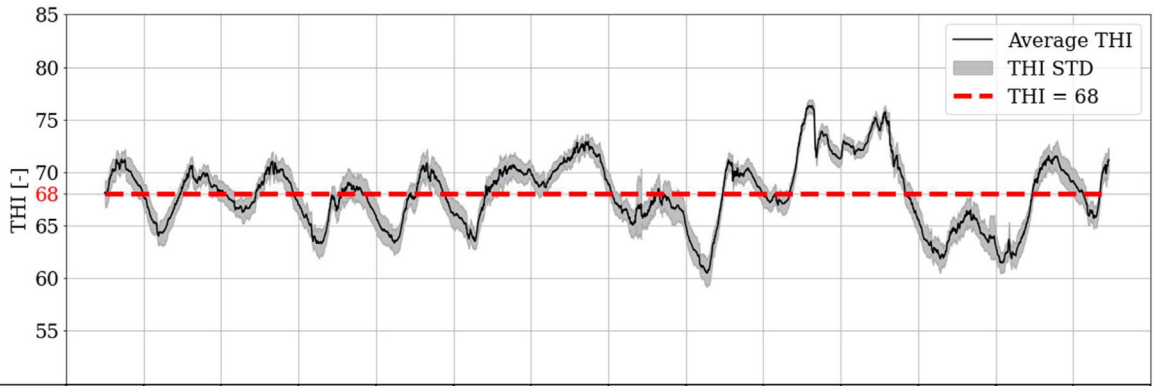
RH = the relative humidity, %

Although each cow has a unique tolerance to warm environmental conditions, it has been reported that dairy cows exhibit early signs of heat stress and experience reduced milk production yield at a THI threshold level of around 68 (Ravagnolo & Misztal, 2002). During the experimental period, the THI level exceeded the pre-determined heat stress threshold of 68 about 58% of the time, as shown in Fig. 5.

2.5. *Evaluation methodologies*

2.5.1. *Paired T-test*

The data sample size of the dairy cows under study was justified by evaluating a statistical power using a paired t-test (Dupont & Plummer, 1990), as presented in Equation (2). Given the significance level and sample size, this power analysis provided the statistical power between two provided sample pools. The higher the power, the more distinction exists between the two compared samples (EBT and CBT in this study), increasing the confidence level of further data analysis.



DOY	231	232	233	234	235	236	237	238	239	240	241	242	243	244
T [°C] (HI/LO)	25/19	23/18	26/17	27/12	28/14	28/15	26/19	24/17	25/13	28/23	27/20	25/17	29/16	29/17
Prec. [mm]	0	2.03	2.79	0	0	0	50.29	1.02	0	11.43	1.02	0	0	0

Fig. 5 – Summary of outside weather and microclimate conditions. (STD = Standard Deviation; Prec. = Precipitation).

$$power = \varphi\left[\delta\sqrt{n} - \left(\frac{\sigma_0}{\sigma_a}\right)z_{\alpha/2}\right] + \varphi\left[-\delta\sqrt{n} - \left(\frac{\sigma_0}{\sigma_a}\right)z_{\alpha/2}\right] \quad (2)$$

where

α = type I error probability for a two-sided test

δ = mean difference

σ_0 = null standard deviation

σ_a = alternative standard deviation

z_a = the critical value that is exceeds by a standard normal variable with probability α

n = number of samples.

2.5.2. *Bland–Altman Plot*

A Bland–Altman Plot is a data-plotting method used to investigate the agreement between two datasets and identify potential bias (Altman & Bland, 1983). In this study, the relationship between measured CBT and EBT was represented by the Bland–Altman method, with the difference between CBT and EBT plotted against EBT and THI levels. This method can provide any bias that might be caused by an internal body-temperature level or by external environmental conditions.

2.5.3. *Temperature gradient*

The temperature gradient was calculated using Equation (3). Each dairy cow produces a unique body temperature profile and physiological response to heat stress. The rate of temperature change can be provided by the temperature profile and its trend agreement between CBT and EBT. In this study, a time length of 12 h is considered the optimal length of time in which to achieve a meaningful gradient plot due to the diurnal nature of the daily cycle.

$$\Delta T = \frac{dT}{dt} \quad (3)$$

where

ΔT = temperature difference, °C.

T = temperature, °C

t = time, hr.

2.5.4. *Regression data fitting*

Various regression models and a simple offset approach were tested in order to develop a model that could capture the CBT trend based on the corresponding EBT trend. Because a simple offset approach adds the average difference that exists between a CBT and an EBT to the EBT, a single-variable linear regression (LR) model (Equation (4)), as well as single variable polynomial regression (PR) model (Equation (5)), were implemented to capture the relationship between CBT and EBT. As presented in Equation (6), a second-order bivariable regression model, a multivariate PR model (Sinha, 2013) was used as a means of considering variations in temperature and including contextual variables such as THI, Time of the Day (TOD), DIM, PAR, and DMAVG. This study considered 10th-order multivariate PR models because all the calculated RMSE evaluations converged after the 10th order.

$$y = \beta_0 + \beta_1 x \quad (4)$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 \quad (5)$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 \quad (6)$$

where

y = predicted value

x_i = input values

β_i = regression parameters

$$Root\ Mean\ Square\ Error = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (7)$$

where

y_i = predicted value

$\hat{y}_i$ = true value

n = number of samples.

2.5.5. *Peak-to-peak comparison*

Determining whether there is any lag between EBT or body temperature response to environmental changes is important.

Accordingly, the daily local maxima for the gradient values of both datasets were identified, and the time differences were compared. The local maxima were determined by a simple comparison of neighbouring values, and a data distance threshold of 12 h was apparently sufficient for detecting the maxima for each day.

3. Results and discussion

3.1. Temperature data summary

Over the course of the two-week experimental period, the ear tag devices collected around 20,000 pairs of CBT and EBT datasets, with the amount of data obtained from each cow ranging from 1000 to 4000. The average temperature readings for CBT and EBT were 38.77 °C and 38.09 °C, respectively, with an average difference of 0.68 ± 0.35 °C (as summarised in Table 2). The temperature spreads recorded at both temperature sources and the standard deviations were slightly higher for the EBT. It should be noted that cows #1, #2, and #7 produced the most milk and recorded higher CBT and EBT averages than did the other cows, reinforcing the observations that higher lactating cows generate more internal body heat and are thus more susceptible to heat stress (Kadzere et al., 2002). Statistical power was calculated for each cow and collectively for all the cows based on the number of collected data. A statistical power of 0.98 was calculated, indicating that the confidence levels concerning the CBT and EBT data obtained from the seven studied cows could be drawn with high confidence.

3.2. CBT vs. EBT

Figure 6 shows the relationship between the CBT and EBT registered by each cow. The x-axis of each graph tracks the EBT, and the y-axis represents the CBT. The polynomial regression line is also provided on each graph as a red curve. Although the relationship between CBT and EBT is primarily linear and can be captured using a linear regression line (Chung et al., 2020), the polynomial curve was used to capture the trailing end of the data where temperature leaks exist (wide EBT spread on the same CBT levels). This is likely because the ear base is more fully exposed to external conditions and is thus more prone to being cooled by water from drinking troughs and to low THI levels than is the vaginal zone. However, it should also be noted that temperature leaks were consistently more common at a lower temperature than at a higher CBT, due to vasodilation which would open the

Table 2 – Average body temperature with standard deviation, average difference between CBT and EBT, and statistical power per each cow and in total.

	Cow #1	Cow #2	Cow #3	Cow #4	Cow #5	Cow #6	Cow #7	Total
CBT [°C]	39.04	38.76	38.63	38.66	38.89	38.73	38.88	38.77
	±0.37	±0.22	±0.23	±0.17	±0.24	±0.20	±0.22	±0.23
EBT [°C]	38.50	37.99	38.23	37.82	37.86	37.88	38.22	38.09
	±0.42	±0.31	±0.28	±0.34	±0.38	±0.32	±0.37	±0.34
CBT-EBT [°C]	0.54	0.76	0.32	0.82	0.96	0.84	0.66	0.68
	±0.26	±0.27	±0.19	±0.30	±0.32	±0.29	±0.32	±0.35
Power	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98

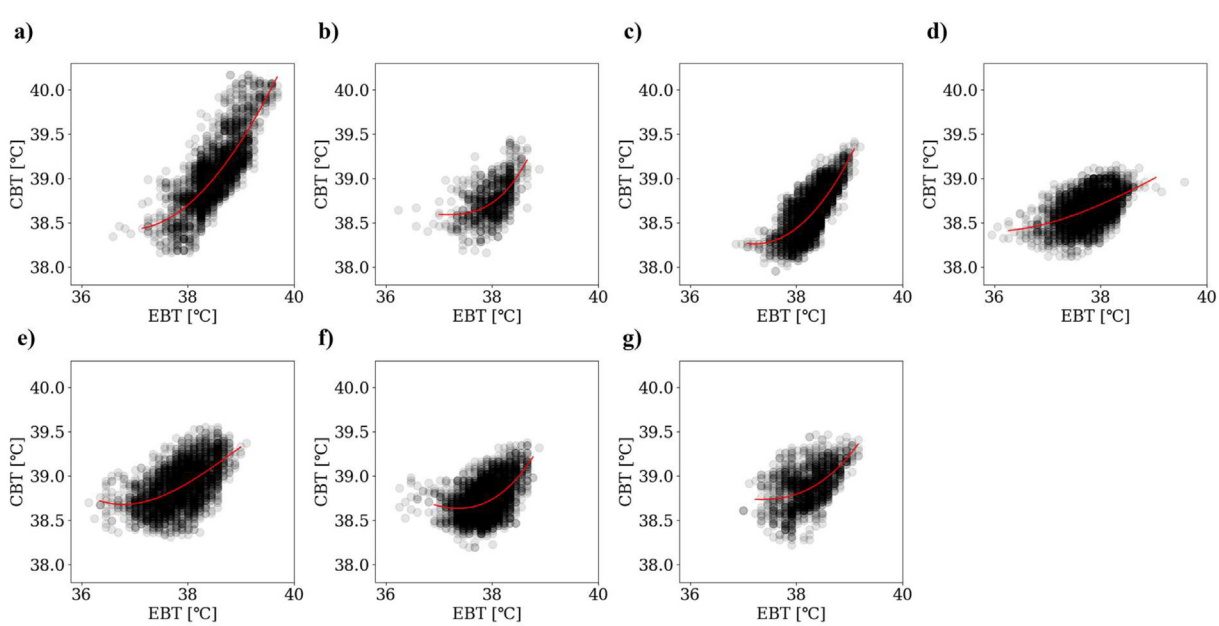


Fig. 6 – XY-plot of CBT and EBT readings with trend line for each cattle. a) to g) is Cow #1 ~ #7, respectively.

blood vessels near the skin surface under heat stress conditions, including the area near the ears. Throughout the experimental period, the average in-barn THI was 68.41, which is over the THI threshold of 68, implying that the summer weather in Wisconsin during the test period was relatively moderate. Given this mild climate, most of the test cows achieved thermal equilibrium through homeostasis. Only Cow #1 (Fig. 6a) experienced detectable heat stress, with her core body temperature climbing to above 40 °C.

While each cow tested during the study was, as stated previously, physically unique, and thus likely to respond at least slightly differently to the same given environmental conditions. Nonetheless, all the cows collectively showed similar overall temperature patterns, and it was therefore crucial to examine the combined data to assess the degree of correlation between CBT and EBT. Figure 7 shows the Bland–Altman plots of the difference between CBT and EBT against the EBT and THI trends for all data. Against the EBT trend (as indicated in Fig. 7a) the difference between the CBT and EBT diminished and became closer to the average difference, of the higher EBT, an outcome which is similar to the conclusion reached by Chung et al. (2020). Clearly, the trend indicates the physiological reaction under heat stress due to cutaneous vasodilation that allows for the necessary redirection of blood flow toward the surface of the skin, where the biosensors are located, for heat dissipation. A similar trend can be observed in the Bland–Altman plot based on the THI level. The occurrences of greater differences between EBT and the corresponding CBT are more frequent at the lower THI levels. Again, this indicates that at a higher THI level and a higher CBT, the EBT becomes more fully insulated from the external conditions and thus represents the true value without any temperature leak. This suggests that when using EBT as a tool to represent CBT during periods of heat stress, data collected at temperatures above 37 °C and a THI level of 68 should be more precise and more reliable.

3.3. *Average temperature trends*

Figure 8a shows the difference between the hourly-averaged CBT and EBT readings throughout the experiment. As can be seen, the EBT follows the same general trend as the CBT. During the experiment, the THI inside the barn was at its highest (rising above the heat threshold of 68) from Day 240–242. Nevertheless, the CBT and EBT readings agreed and rose and fell along with the THI. Although there were occasional and significant spikes in the implant's temperature readings, they consistently occurred during the morning milking time (4 AM). While the THI was not measured when the cows were in the milking parlour or the holding area, the THI would most likely have been much lower throughout the night, when no animals were present and any sudden change in external air conditions would have been likely to impact the EBT. Figure 8b shows the temperature gradient for CBT and EBT based on time. A 12-h time window was selected for calculating temperature gradients because, given the diurnal nature of the day and temperature cycle, 12-h windows produce a smooth curve with gradients for both the CBT and EBT showing comparable trends.

Figure 9 depicts the side-by-side comparison of hourly average EBT and THI. Although THI is a good indicator of body temperature trends, it is not, for the most part, perfect. During the high-THI period, the THI level rose and fell, and the body temperature followed this trend, indicating that a lag time existed between the rise in THI and the cow's physiological response to that rise. Pick-to-pick correlations were analysed with the local maxima for each data trend used to determine the exact degree of lag between the CBT, EBT, and THI readings. This analysis showed that virtually no lag occurred between the two temperature readings (with nearly 0 h), while the temperature peaks occurred on average about 3.34 h after the THI peaks. However, the time lag between the two variables fluctuated from day to day, ranging from 2 to 6+ hours. This finding corresponds with the trend described by Atkins

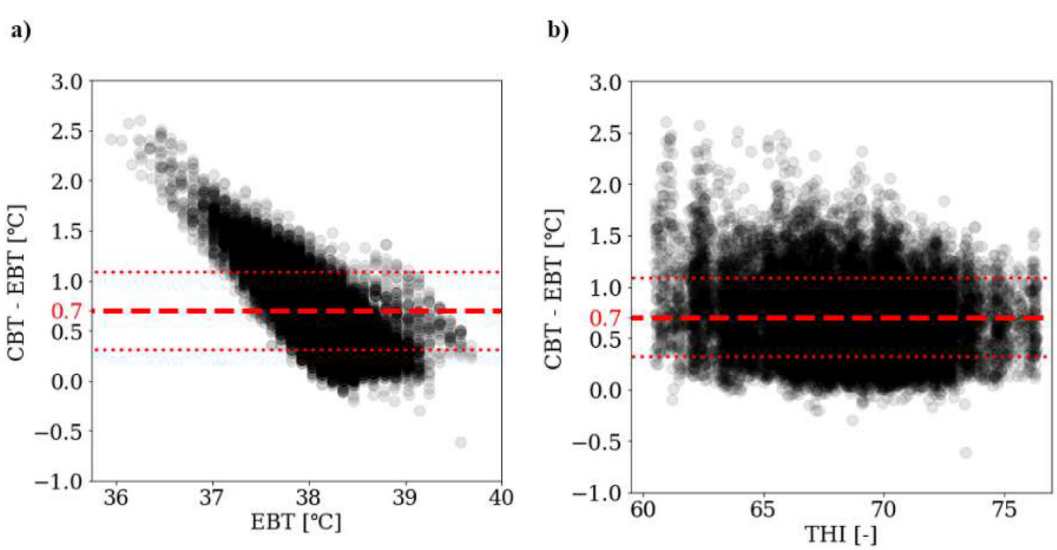


Fig. 7 – Bland–Altman plots: a) the difference between CBT and EBT based on EBT trend; b) the difference between CBT and EBT based on THI trend. Dash lines indicate average difference and dotted lines indicate standard deviations.

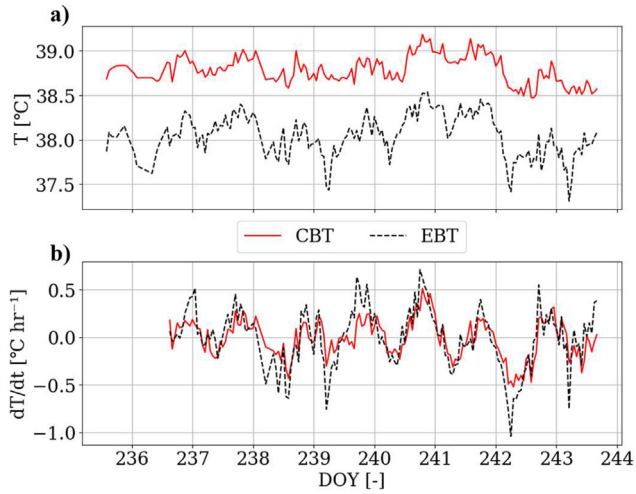


Fig. 8 – Timeseries plot of averaged CBT and EBT trends of all cattle. a) temperature plot; b) temperature gradient plot of 12-h time window.

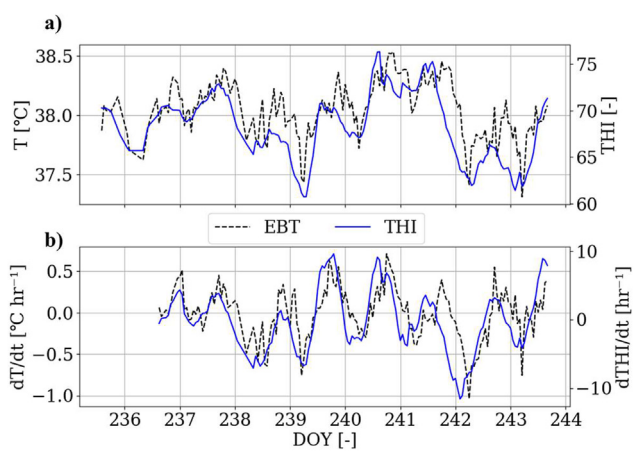


Fig. 9 – Timeseries plot of averaged THI and EBT trends of all cattle: a) absolute values; b) gradient plot of 12-h time window.

et al. (2018), in which the body temperature of the study's test cows remained elevated despite the decreased THI level in the evening. Using THI to anticipate heat stress level could be a viable option. However, the uncertainty related to the cow's body temperature response delay is far from being precise, especially in larger commercial dairy barns where microclimate conditions are not uniform.

3.4. *Regression data fitting*

For EBT to stand alone, a regression model that can predict CBT based on EBT should be developed based on the entire dataset. Figure 10 shows three single-variable regression models capable of capturing accurate CBT levels based on a presented EBT of all seven cows. Overall, the CBT consistently correlated positively with the EBT, and the three models, simple offset based on the average difference (Offset), linear regression (LR), and second-order polynomial regression (PR)

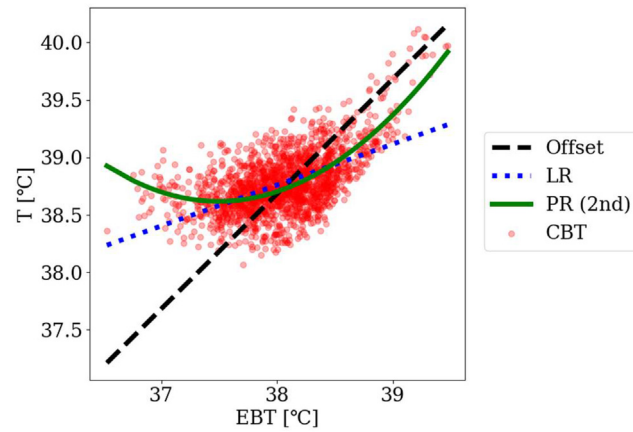


Fig. 10 – CBT vs. simple offset data fitting and single variable regression models. (LR = Linear Regression; PR = Polynomial Regression).

captured the CBT with average errors of 0.32, 0.21, and 0.19, respectively. The PR model works best with a single parameter (EBT), most likely because the EBT behaves with greater accuracy than does the CBT.

The EBT alone can capture the CBT with a moderately reasonable degree of confidence. However, a cow's body temperature is a complex amalgam of many variables involving the body's interior and external conditions. Consequently, multivariate PR was also used to improve the regression accuracy. Figure 11 shows the prediction performance of multivariate PR models (which were fitted based on the entire EBT dataset), for predicting, for example, the CBT of Cow #1. In addition to the EBT, the three presented models considered either the THI or TOD as additional variables, or both the THI and TOD. The result showed average errors of 0.19, 0.20, and 0.17, respectively, for those three models. Considering at least one more variable drastically improved the regression model's ability to fit the CBT trends and considering both THI and TOD further improved the accuracy. The overall performance and the individual performances of each cow, as presented by the regression models, are summarised in Table 3.

The CBT, EBT, THI, and TOD are treated as independent variables in this study. Adding certain variables related to a cow's health or production level can potentially further improve the regression model's performance. The lists of RMSE scores shown in Table 4 are derived from a different combination of such additional parameters as DIM, PAR, DCC, and DMAVG. The models that included DMAVG consistently performed the best in predicting the overall CBT (with an average error as low as 0.11 °C). This finding indicated that a meaningful correlation exists between the amount of milk each animal produced and that animal's body temperature trend.

Regression models offer excellent ways to evaluate the relationship between the two temperature trends (CBT and EBT). By taking a machine learning approach to the multivariate polynomial regression mode, many different variables (even apparently unrelated variables) can be added to provide more information concerning the animals and the

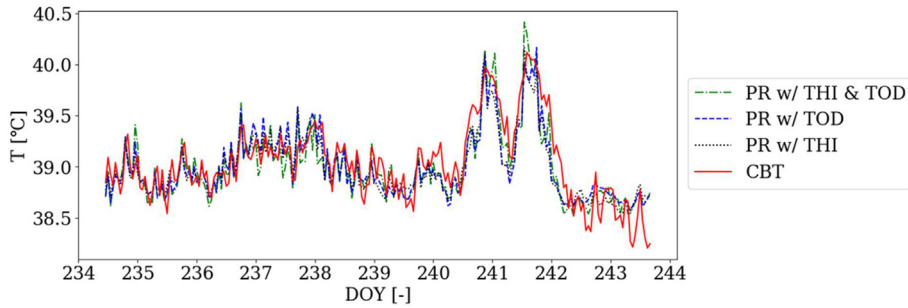


Fig. 11 – Demonstration of multivariate polynomial regression prediction against CBT of Cow #1.

Table 3 – RMSE scores for different temperature data fitting models.

Models [°C]	Cow #1	Cow #2	Cow #3	Cow #4	Cow #5	Cow #6	Cow #7	Total
Offset	0.25	0.27	0.37	0.26	0.44	0.27	0.26	0.32
LR	0.27	0.18	0.28	0.12	0.25	0.15	0.17	0.21
PR (2nd)	0.23	0.18	0.25	0.13	0.25	0.15	0.18	0.20
PR w/THI	0.19	0.19	0.24	0.13	0.24	0.16	0.18	0.19
PR w/TOD	0.20	0.18	0.23	0.12	0.24	0.14	0.17	0.20
PR w/THI & TOD	0.17	0.16	0.22	0.12	0.23	0.14	0.17	0.17

Table 4 – The full summary of RMSE scores for multivariate polynomial regression model on top of EBT, THI, and TOD.

Additional Parameters	Total RMSE [°C]	Additional Parameters	Total RMSE [°C]
DIM	0.17	PAR, DCC	0.13
PAR	0.17	PAR, DMAVG	0.12
DCC	0.14	DCC, DMAVG	0.12
DMAVG	0.13	DIM, PAR, DCC	0.12
DIM, PAR	0.17	DIM, PAR, DMAVG	0.11
DIM, DCC	0.12	DIM, DCC, DMAVG	0.11
DIM, DMAVG	0.11	DIM, PAR, DCC, DMAVG	0.11

environment and thus further improve the regression model's accuracy. The model used in this study considered all the salient temporal (TOD), physiological (CBT and EBT), and production (DIM, PAR, DCC, DMAVG) variables to yield an accurate prediction with a highly acceptable error and two weeks' worth of datasets. With more variables and a larger dataset, the method used in this study could yield even better results, and the EBT could become a crucial component for future heat stress detection systems, dramatically improving the methods dairy farmers use to manage the conditions inside their barns to maintain their animals' well-being.

3.5. *Real-world applications of EBT readings*

For real-world applications and engineering aspects, spacio-temporal data sets are required to provide cool air and/or spray water in real-time to the cows experiencing heat stress. Therefore, a closed-loop system, as presented in Fig. 1, is required to determine if fans and soakers should be operated and at what level. In addition to the runtime precision control, the design-time optimization needed to determine the numbers and locations of these cooling methods are also necessary. The system based on physiological (e.g., EBT or CBT), behavioural (e.g., GPS locations, resting, standing, bunching, and feeding), and environmental (e.g., temperature, humidity, and wind speed) data sets should allow engineers and producers to automate, monitor, and make real-time adjustments by bringing ventilation and cooling into a single decision-making unit. In a unique way, the cloud-based infrastructure, integrated with IoT units, should be capable of controlling variable-frequency-drive (VFD) fans and/or mist and fogging nozzles. In this regard, a few commercial products are already under development and in use in recent years, such as Smartbow® (Zoetis Inc., Parsippany–Troy Hills, NJ, USA) for herd positioning and monitoring and DairyBOS® (VES-Artex, Chippewa Falls, WI, USA) for controlling VFD fans by a real-time, wireless microenvironmental data set. Likewise, the EBT data sets (physiological), along with behavioural and environmental data sets, can be further integrated into a smart decision-making package to improve productivity, welfare, and energy and water efficiency in modern dairy barns.

4. Conclusions

In lieu of the traditional, more invasive ways of measuring core body temperature, each cow's body temperature was measured by a sensor attached to the animal's ear tag with a biosensor. Results showed that EBT can comparably represent the CBT trends with almost no time lag. On average, the difference between a test cow's EBT and a corresponding CBT was about 0.68 ± 0.35 °C, which is the deviation to be expected when comparing temperatures taken from a location on an animal's core to temperatures taken from the subcutaneous biosensors. The relationship between the EBT and the CBT of each cow tested in this study varied, and therefore, instead of relying on an animal's absolute temperature

value, either the rate of temperature change, or the relative temperature difference should be considered when using EBT data to detect the subtle body-temperature increases that signal the onset of heat stress. Data-fitting models can further adjust the recorded EBT readings and accurately predict the CBT within 0.2 °C, given that the body temperature of each cow will invariably differ by at least that much because of variations in physiological parameters and milk yield. As expected, a consistent time lag existed between an increase in the environmental temperature and a subsequent rise in CBT; moreover, the EBT measurements tended to become increasingly precise as the THI exceeds 68, the heat stress threshold, due to vasodilation. Overall, the outcomes obtained during this study indicate that using an ear-tag sensor to measure EBT could effectively serve as an alternative to the invasive CBT-measurement approach.

The EBT of each animal used in this study was also successfully measured and observed in real-time by means of a custom-designed ear-tag device that can wirelessly transmit EBT to any IoT system. The lightweight and compact design of this ear tag was made possible by recent advancements in wireless battery recharging capabilities and electronics that make power consumption more efficient. Given this study's findings, ear-tag-based body temperature monitoring should become a crucial part of precision livestock agriculture, one that focuses on physiological feedback to manage an animal's environment and health and maximise the animal's well-being as well as productivity more efficiently.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We acknowledge the contribution of Professor Sabrina H. Brounts to consulting with the biosensor implantation site and performing the injection procedure. We would also like to thank Mr. Xi Zhang and Mr. Trey Sandiford for assisting with the animal handing. This research was supported by USDA National Institute Food and Agriculture (Joint NSF Cyber-Physical Systems) under Grant No. 67021–34036.

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